



DP & 0/1 Knapsack



Ek dum Bachchon Jaisi Simple Language Mein! (Hinglish)

1. Dynamic Programming (DP) Kya Hai? 🤔

Bohot se log DP ka naam sunkar darr jate hain, jaisa video mein bataya gaya. Par asal mein yeh koi bhoot nahi hai!

Simple funda: Jo kaam tumne ek baar kar liya hai, uska result yaad rakh lo (Memory mein save kar lo). Taki future mein agar wahi same kaam aaye, toh wapas mehnat na karni pade. Is "yaad rakhne" ki trick ko **Memoization** kehte hain.

Example: Agar kisine poocha $2+2$ kya hai? Tumne count kiya "4". Agli baar koi wapas poochega, toh wapas ungli pe count nahi karoge, direct "4" bologi!

2. 0/1 Knapsack Problem Kya Bimari Hai? 🎒

Maan lo tum ek dukaan mein gaye ho aur tumhare paas ek **Jhola (Bag)** hai. Jhole ki ek limit (Capacity) hai, usse zyada weight daloge toh bag phat jayega.

- **0 ka matlab:** Item ko jhole mein *NAHI* daalna (Chhod do).
- **1 ka matlab:** Item ko jhole mein *DAAL LENA*.

Tumhara main goal hai: **Bag full hone se pehle, sabse mehengi (zyada profit wali) cheezein pick karna.**

Items:



+

Tumhara Bag



3. Soubhik's Real Life Example! 💪💡

Soubhik, DP and Knapsack ko apni daily life aur memory se link karke dekho:

- **Gym & High-Protein Diet (Protein Knapsack):** Tumhara pet ek Knapsack hai jiski maximum calorie limit (Capacity) hai. Items hain *Whey Protein (ATOM)*, *Chicken Keema*, *Soyabean*. Tumhe wahi item khana hai (1) jo tumhe sabse zyada Protein (Profit) de, lekin daily Calorie limit cross na ho!
- **Decorative Brass Lights Business:** Tumhare paas shipment aayi hai. Truck mein limited jagah (weight limit) hai. Tum wahi brass lights truck mein load karoge (1 - Include) jisme margin zyada ho, aur baki lights ko chhod doge (0 - Exclude).
- **UrbanPro Python Classes:** Jab tum apne students ko DP padhaoge, toh unko yahi Zomato order ya Gym diet wala example dena. Bacche DP ka concept life-long nahi bhulenge!

4. Code Kaise Kaam Karta Hai? 💻

Har item ke paas khade hokar khud se poocho: "*Kya isko lene se mera overall profit badhega ya nahi?*" Aur result table (DP Memo) mein save kar lo.

```
# Python Code - 0/1 Knapsack (Memoization)
```

```
def knapsack(wt, profit, W, n, memo):
    # Agar items khatam ya bag ki capacity khatam
    if n == 0 or W == 0:
        return 0

    # DP Magic: Agar already calculated hai, directly use karo
    if memo[n][W] != -1:
        return memo[n][W]

    # Agar item ka weight bag limit (W) se bada hai -> Exclude (0)
    if wt[n-1] > W:
        memo[n][W] = knapsack(wt, profit, W, n-1, memo)
        return memo[n][W]

    # 2 Choices: LENA HAI (1) ya NAHI LENA HAI (0)
    profit_include = profit[n-1] + knapsack(wt, profit, W - wt[n-1],
n-1, memo)
    profit_exclude = knapsack(wt, profit, W, n-1, memo)

    # Dono mein se jo MAX profit de, use DP table me save kar lo
    memo[n][W] = max(profit_include, profit_exclude)

    return memo[n][W]
```

5. Important Points

- Bina DP ke (sirf recursion), yeh code har raste ko baar-baar calculate karega jo ki bohot slow hota hai (Exponential Time).
- Har step pe hum ek "If-Else" jaisa decision lete hain (Include ya Exclude).
- Ek baar Knapsack samajh gaye, toh bohot saari aisi similar DP problems (subset sum, etc.) solve kar sakte ho!